EasyA Team Maths Challenge

SUGGESTED ANSWERS

Points to note:

- To earn full marks, students must show their logical progression on all of their answers. A rule of thumb is, a Part A (GCSE / Easier AS Level Questions) answer must be written in a way readable and accessible to GCSE / AS Level students. Part B to AS / A Level students, Part C to A-Level students.
- For each question, there will be remarks commenting on the general performance.
- The questions are written by Ryan Lam, with help from other EasyA Super Tutors mentioned at the later parts.
- The following answers are suggestive and could be used as a reference only. Any correct work with clear logical progression should be awarded full marks as well.

Section A: GCSE Questions (5 Marks Each)

1. Question 1

Show that in any year, the 1st of April is always on the same day of the week as the 1st of July.

Solution:

Between the 1st of April and 1st of June, there are $30 + 31 + 30 = 91$ days.

Two days have the same day of the week if and only if they are 7, 14, 21… days apart (or if the number of days between the two days is divisible by 7), start and end inclusive.

91 is divisible by 7, so they are on the same day of the week as required.

Remarks: Satisfactory performance. Some students have not justified why will there be 91 days between the three months, some mistakenly write 93 days.

2. Question 2

You have 5 gold coins (Call then coin A, B, C, D, E if you want) identical in appearance. One of them is a fake coin. You are given a truthful balance as shown in the figure. You know the fake coin is heavier, you want to know which coin is fake.

Describe a way such that you can know the answer using the balance exactly 2 times.

Solution:

On one side of the coin, put coins A and B, put coins C and D on the other side.

If the two sides are balanced, then E is the fake coin.

If one side is lighter, say the side with AB, then weigh A against B.

The lighter one is the fake coin since we know at least one of A, B must be normal.

Remarks: There are lots of different answers to this puzzle, full marks should be given to answers that make complete logical sense.

Excellent performance for this question in general.

3. Question 3

Find 5 numbers where, when considered together, have a mean of 18, median of 20, and mode of 22. Explain your answer.

[Hint: There is more than one choice, any choice will be acceptable.]

Solution:

Name the number a, b.

Now the mode is 22, so we know there are at least two "22"s in the set of 5 numbers.

Arrange the number into Descending order, since 20 is the median, we know the first three numbers are 22, 22, 20.

We have the five numbers summed up to $18 \times 5 = 90$. So the remaining two numbers must satisfy $a + b = 90 - 22 - 22 - 20 = 26$

One such choice is $a = 12, b = 14$, so the five numbers are 12,14,20,22,22

Remarks: Answer is correct if and only if the numbers a, b are strictly between 6 to 20 and $a = b = 13$ is NOT acceptable.

4. Question 4

From 7 A.M. to 7 P.M., the hour hand of a clock makes one full revolution. How many times do the minute hand and the hour hand overlap? Explain.

What is the time where it happens, and it is closest to 11:30 A.M.?

Solution:

The first overlapping happens after 7 A.M. and before 8 A.M.

When the hour hand made one full revolution around the clock, the minute hand made 12 revolutions.

That is, when the minute hand make one complete revolution, the hour hand completes 1/12 revolution, then it overlaps. Note that each overlapping takes slightly more than an hour

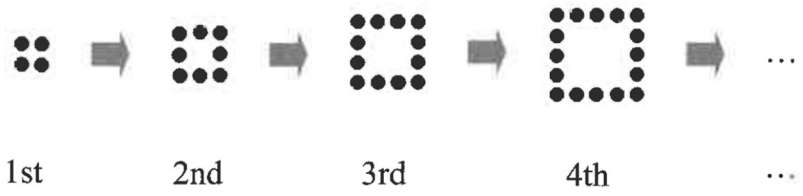
Hence there are 11 times of overlapping, and the closest to 11:30 A.M is at 12 noon.

Remarks: Again, there are a lot of different answers to justify. One common alternate way is just to draw all the cases. Satisfactory

performance in this question although some teams haven't consider all cases.

5. Question 5

Consider the following pattern with some dots:



Is there a pattern consisting of 82 dots? Explain.

Solution:

Note that each pattern is formed by adding 4 dots, **OR**

The nth pattern can be found by making a square with $(n+1)$ dots, then removing a square of $(n-1)$ dots: so the total number of dots: $(n+1)^2 - (n-1)^2 = 4n$ dots.

So the number of dots in any pattern is divisible by 4.

But 82 is not divisible by 4 so there is no such pattern.

Remark: Most teams get 1 mark deducted because the justification of each pattern contains 4n dots is not clear.

6. Question 6

Alex is 15% taller than Ben, while Charlies is 12% shorter than Alex.

It is given that Ben is 160 cm tall.

Who is the tallest among the three people? Explain.

Solution:

Alex's Height $= 160 \times (1 + 15\%) = 184$ cm

Charles' Height $= 184 \times (1 - 12\%) = 161.92$ cm

$184 > 161.92 > 160$ so Alex is the tallest.

OR, since Alex is Taller than Charles and, Alex is taller than Ben, Alex is taller than any one of them, so he is the tallest.

Remarks: I want to ask about shortest originally, but this delivers a nice shortcut way of looking at the problem. Both ways are of course acceptable. Excellent performance on this question.

7. Question 7

a, b are positive numbers such that

$$\frac{6b+5a}{9a-5b}=6.$$

Work out $\sqrt{a}:\sqrt{b}$.

Solution:

$$6b+5a=6(9a-5b)\Leftrightarrow 6b+5a=54a-30b\Leftrightarrow 36b=49a\ \Leftrightarrow \frac{a}{b}=\frac{36}{49}$$

For $\sqrt{a}$ (resp. $\sqrt{b}$) to be defined a, b must be both positive (optional.), so

$$\frac{\sqrt{a}}{\sqrt{b}}=\sqrt{\frac{a}{b}}=\frac{6}{7}\Leftrightarrow \sqrt{a}:\sqrt{b}=6:7$$

Remarks: I use $\Leftrightarrow$ for saving space, but usually, secondary school students just open a new line. Poor performance on this question. Most of the teams incorrectly concluded the answer as $7:6$.

8. Question 8

The stem-and-leaf diagram below shows the distribution of exam results of some students in a class.

Stem (tens)	Leaf (units)
6	1 1 1 3 4 6 8 9 9
7	a 7 7 8
8	1 b

It is given that the mean and the range of the above distribution is 70 and 22 respectively. Find a, b.

Solution:

There are 15 students in this class, there is two marks that we don't know (which is $70+a$ and $80+b$ marks). We can see the other 13 students' marks added up to 895, so, since the mean is 70:

$895 + (70 + a) + (80 + b) = \ 70 \times 15 \Leftrightarrow 1045 + a + b = 1050 \Leftrightarrow a + b = 5.$

Now the range of the distribution is $(80 + b) - (61) = 22$ so $b = 3$, and hence $a = 2$.

Remarks: Some students use a, b instead of $80 + b, 70 + a$, which is also fine.

9. Question 9

The circle C is the circle that passes through A: $(-6,5)$ and have centre B: $(2, -1)$. Find the equation of C.

Given the point D: $(-3, 11)$ that lies outside the circle.

H is a point on the circle where it is furthest from D.

What is the relationship between points B, D, H? Find the equation of the straight line which passes through D and H. [Hint: You might find drawing a diagram here helpful.]

Solution:

The radius of the circle C is: $\sqrt{(2-(-6))^2 + (5-(-1))^2} = 10$

So, the equation of the circle is $(x-2)^2 + (y+1)^2 = 100$

The points B, D, H are collinear (no explanation needed).

So the slope of the line connecting (B), D, H is: $\frac{11-(-1)}{-3-2} = -\frac{12}{5}$

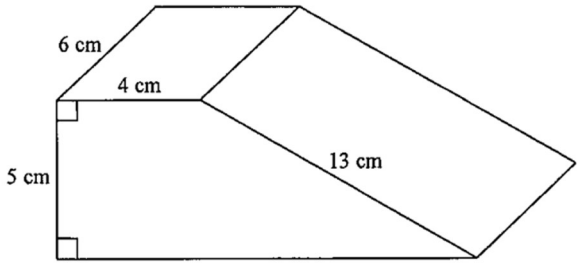
The equation of the line is $y = -\frac{12}{5}x + \frac{19}{5}$ (or any equivalent answer)

Remark: A lot of the scripts have not completed the second question.

Any words with the same meaning as "collinear" (e.g. "lie on the same straight line") would also earn 1 mark.

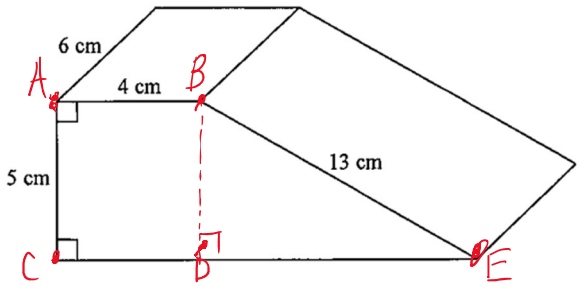
10. Question 10

Find the volume of the right prism shown:



Solution:

This is a prism with the base as the trapezium facing towards us. Draw a suitable vertical line at the base and name accordingly:



$ABCD$ is a rectangle so $BD = 5$ cm, now ΔBDE is a right-angled triangle, so

by Pythagoras Theorem: $DE = \sqrt{13^2 - 5^2} = 12$ cm,

hence the area of the base is $Area\ of\ ABCD + Area\ of\ \Delta BDE = AB \times AC + \frac{BD \times DE}{2} = 4 \times 5 + \frac{5 \times 12}{2} = 50\ cm^2$.

Hence the total volume of the prism is $50 \times 6 = 300 cm^3$ by multiplying the base area with height.

Remarks: Note that the first two marks are fully awarded for any correct methods. Fair performance on this question, one mark is usually deducted for unclear use of Pythagoras theorem.

[END OF SECTION A]

Section B: Easier AS Level / A Level Questions (10 Marks Each)

11. Question 11

You have a rope of length 100 m, you cut it into two parts.

You make a square and a circle with the two parts respectively.

Given the square and the circle has the same area, what is the length of the first part of the rope?

[Hint: Your answer should be in the form of

$$\frac{a}{b\sqrt{\pi}+c}$$

, where a, b, c are constants to be determined.]1, 3, 5, 4, 2

Solution:

Let the required length be x cm long, then $100-x$ cm was used to make a circle.

The side length of the square is $x \div 4$ cm. Now, if the radius of the circle is r cm, we have $2\pi r = 100 - x$, so $r = \frac{100-x}{2\pi}$.

Now the area of the square and the circle are the same, so:

$$d^2 = \pi r^2 \Leftrightarrow \left(\frac{x}{4}\right)^2 = \pi\left(\frac{100-x}{2\pi}\right)^2 \Leftrightarrow \left(\frac{x}{4}\right)^2 = \left(\frac{\sqrt{\pi}(100-x)}{2\pi}\right)^2$$

$$\Leftrightarrow \left(\frac{x}{4}\right)^2 - \left(\frac{\sqrt{\pi}(100-x)}{2\pi}\right)^2 = 0$$

$$\Leftrightarrow \left(\frac{x}{4} - \frac{\sqrt{\pi}(100-x)}{2\pi}\right)\left(\frac{x}{4} + \frac{\sqrt{\pi}(100-x)}{2\pi}\right) = 0$$

$$\Leftrightarrow x = \pm\frac{4\sqrt{\pi}(100-x)}{2\pi} \Leftrightarrow 2\pi x = \pm 4\sqrt{\pi}100 \mp 4\sqrt{\pi}x$$

$$\Leftrightarrow (2\pi \pm 4\sqrt{\pi})x = \pm 4\sqrt{\pi}100$$

$$x = \pm\frac{4\sqrt{\pi}100}{2\pi \pm 4\sqrt{\pi}} = \pm\frac{400}{2\sqrt{\pi} \pm 4} = \pm\frac{200}{\sqrt{\pi} \pm 2}$$

Note that the negative answer: $x = -\frac{200}{\sqrt{\pi}-2}$ will be rejected.

So $x = \frac{400}{2\sqrt{\pi}+4}$ *or* $\frac{200}{\sqrt{\pi}+2}$ (both are accepted answers.)

Remarks: Any method of simplifying the quadratic equation should be awarded at least 2M if most of the algebraic manipulation makes sense, regardless of the correctness of the answer. It is not advisable to let x be

the circle's length due to the question asking for the length of the first part.

12. Question 12

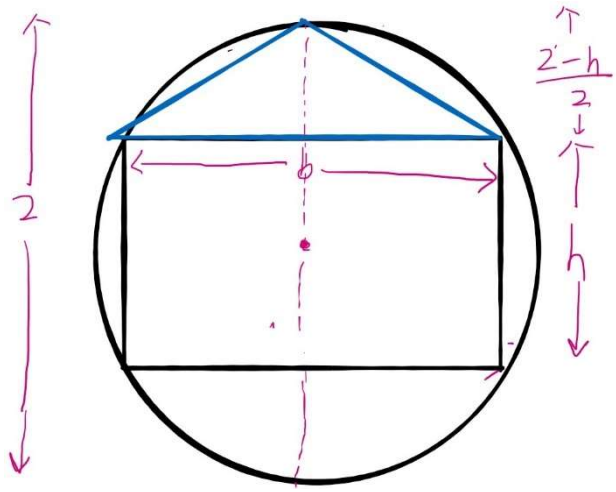
Inscribe a rectangle of base b and height h in a circle of radius 1, and inscribe an isosceles triangle in the region of the circle cut off by one base of the rectangle, with that side of the rectangle as the base of the triangle.

Draw a rough sketch of the situation described above.

For what value of h do the rectangle and triangle have the same area?

Solution:

Attached is the required figure:



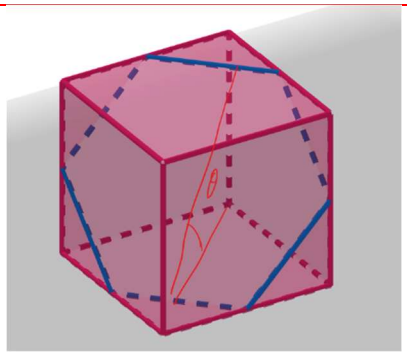
The rectangle's area is bh, triangle's area is $\frac{1}{2} \cdot b \cdot \frac{2-h}{2}$, (the $\frac{2-h}{2}$ comes from the diameter of the circle is 2), equating both sides we have

$$\frac{1}{2} \cdot b \cdot \frac{2-h}{2} = bh \Leftrightarrow \frac{2-h}{4} = h \Leftrightarrow h = \frac{2}{5}$$

Remarks: Generally well done attempt here. Some teams drawn the wrong figure or haven't present all information on the figure, 1 mark will be deducted.

13. Question 13

The following figure is a cube with all faces of square length 1. A hexagon (shown in blue) is formed by taking midpoints of six sides as shown. Let the angle between the base of the cube and the hexagonal surface be θ, work out $\cos\theta$.

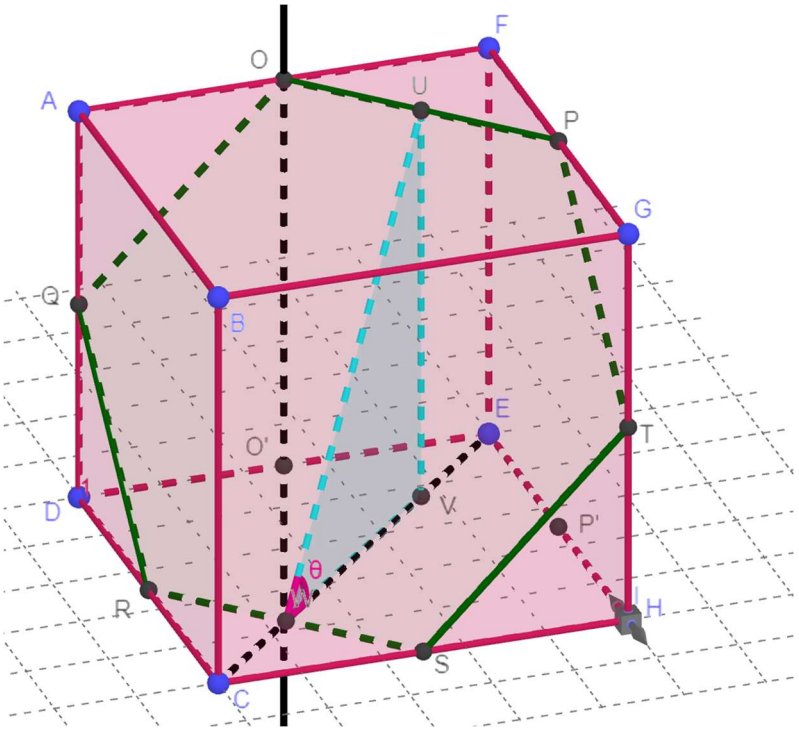


Solution:

We Name our Points as follows:

- The (unit) cube $ABCDEFGH$, the hexagonal surface given $QRSTPO$,
- The midpoint of PO to be U, the midpoint of RS to be W,
- And if we project the line OUP to the base of the square, name the line $O'VP'$.
- Now note the required angle θ is exactly $\angle VWU$, which is convenient because ΔVWU is a right-angled triangle with $\angle UVW = 90^o$.

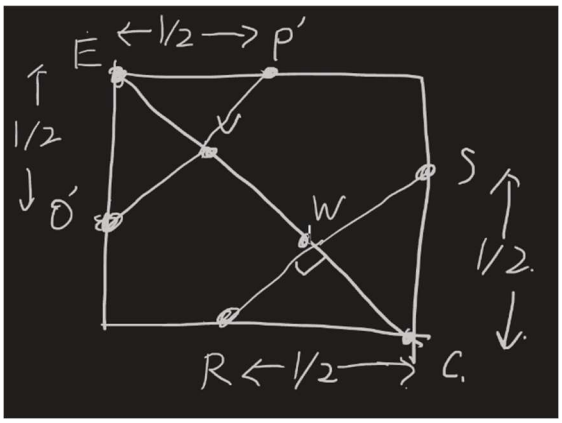
(If you don't like words, feel free to think about the photo below:)



(Note: The line OUP is projected according to the vertical black line in the photo. You can see it interactively here.)

Now, to find $\cos\theta$ we only need to know the adjacent side VW and hypotenuse UW, but now, we only know $UV = 1$ since it is a unit cube.

We can also easily find VW by considering E, V, W, C are collinear and lies on the base of the cube:



By Pythagoras theorem, one has $RS = \frac{\sqrt{2}}{2}$, then W bisects RS (because ΔRSC is an isosceles triangle and WC is the altitude) so $RW = WS = \frac{\sqrt{2}}{4}$. By Pythagoras theorem again on ΔRWC, $WC = \sqrt{RC^2 - RW^2} = \frac{\sqrt{2}}{4}$. The same reasoning shows $EV = \frac{\sqrt{2}}{4}$. So

$$VW = EC - EV - WC = \sqrt{2} - \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4} = \frac{\sqrt{2}}{2}$$

Now we can finally conclude, since $UW = \sqrt{VW^2 + UV^2} = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}$

Hence, $\cos\theta = \frac{VW}{UW} = \frac{\sqrt{2}}{\sqrt{6}} = \frac{\sqrt{3}}{3}$.

Remark: This is a question, again with various approach, some other alternate solution includes working out the plane equation / normal vector of both planes and using formulas to calculate angles between two planes.

Also, the angle in ΔEUW can also be considered, and there are few teams following this approach.

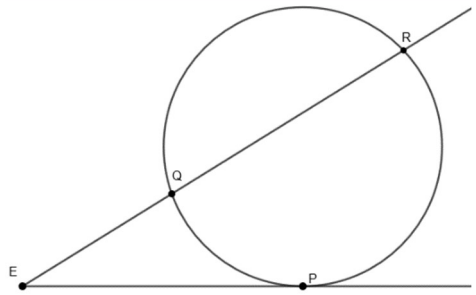
However, I found this solution the most elementary, only requiring the intuitive understanding of angle between two planes, but not any (complicated) formulas.

Generally poor performance in this question, usually because of unclear presentation making the answers very hard for markers to follow.

14. Question 14

The following figure shows a circle passing through three distinct points, P, Q and R. Let E be a point lying outside the circle such that E, Q, R lies on the same straight line, show the following statement:

$QE \times RE = PE^2$ <u>if and only if</u> PE is a tangent to the circle at P.



Solution:

"If" part:

$$QE \cdot RE = PE^2 \quad \text{so} \quad \frac{QE}{PE} = \frac{PE}{RE}$$

Also, $\angle QEP = \angle PER$ (Common Angle)

$\triangle QEP \sim \triangle PER$ (2 Sides corr., inc. $\angle$)

$\angle QPE = \angle PRE$ (corr. $\angle$s, $\triangle$s)

So EP is a tangent to the circle at P. (Converse of $\angle$s in alt. segment)

"Only if" Part:

So EP is a tangent to the circle at P.

$\angle QPE = \angle PRE$ ($\angle$s in alt. segment)

$\angle QEP = \angle PER$ (Common Angle)

$\triangle QEP \sim \triangle PER$ (A.A)

$$\frac{QE}{PE} = \frac{PE}{RE} \quad \text{so} \quad QE \cdot RE = PE^2$$

Remark:

Here are some explanations of the notion:

(2 Sides corr., inc $\angle$) is a reason for two triangles to be similar because there are two sides in correspondence ratio and the included angle of the two side is the same.

(corr. $\angle s$, similar Δ) is given a similar triangle, their corresponding angles are the same.

($\angle s$ in alt. segment) is the claim that A line is a tangent to a circle if and only if the angle in the alternate segment of the circle is the same.

Generally poor performance, because students can not grasp the meaning of "if and only if".

15. Question 15

Find the complex numbers z, w that satisfies the following equation.

$$\begin{cases}(1+i)z+(1-i)w=3-i\\(1-i)z+(1+i)w=7+i\end{cases}$$

Solution:

Let $z=x+yi, w=u+vi$, where x,y,u,v are real numbers, then by substitution:

$$(1+i)(x+yi)+(1-i)(u+vi)=3-i\ldots(1)$$
$$(1-i)(x+yi)+(1+i)(u+vi)=7+i\ldots(2)$$

Expanding and simplifying equations (1) and (2) respectively, we have:

$$(x+u+v-y)+(x-u+v+y)i=3-i$$
$$(x+y+u-v)+(y-x+u+v)i=7+i$$

By Comparing coefficients, we have the simultaneous equations:

$$x-y+u+v=3\ldots(3)$$
$$x+y-u-v=-1\ldots(4)$$
$$x+y+u-v=7\ldots(5)$$
$$-x+y+u+v=1\ldots(6)$$

Now adding $(3),(4),(5),(6)$ together, we have

$$2x+2y+2u+2v=10 \quad \text{hence} \quad x+y+u+v=5$$

Subtracting (7) with $(3),(4),(5),(6)$ respectively we have $x=2,y=1,u=3,v=-1$, so $z=2+i, w=3-i$.

Remarks: Various ways can reach the answer, full marks will be awarded if the logical progression is sound.

Some teams only stated the answers to the question, which will only be awarded 2 marks maximum.

16. Question 16

20 deer live in a forest, out of which 7 are tagged. Later on, 4 random deer of the 20 are captured.

What is the probability that among those 4 captured deer, 2 will be tagged?

Solution:

The required probability is

$$\frac{Number\ of\ Exactly\ 2\ of\ deer\ out\ of\ 4\ will\ be\ captured}{Number\ of\ different\ ways\ that\ 4\ deer\ will\ be\ chosen\ out\ of\ 20}$$

The numerator is $\binom{7}{2}\binom{13}{2}$, this is because out of all the possible ways to choose exactly 2 tagged deer from 4, is to choose the 2 tagged deer from 7, then choose the remaining 2 $(4-2=2)$ among the untagged 13.

The denominator is $\binom{20}{4}$ because we allow every choice of 4 deer.

So the required probability is

$$\frac{C_2^7 C_2^{13}}{C_4^{20}} = \frac{546}{1615}$$

Remark: Note that both notations $\binom{n}{r}, C_r^n$ are interchangeable, as demonstrated in the solution. And on a personal note, I like how this question is deceivingly simple. [P.S. Also, as an ESL speaker, I often forgot the plural of "deer" is still "deer", but not "deers".]

Generally satisfactory performance on this question, the markers would like to see more explanation provided on how to reach the answer.

[END OF SECTION B]

Section C: Harder A Level Questions (15 Marks Each)

I would like to thank Edwin and Jonas, who are both EasyA Tutors, for writing up the solution for this part.

17. Question 17

A sequence of polynomials $(f_0(x), f_1(x), f_2(x) \dots)$ satisfies

$$\frac{d}{dx} f_{n+1}(x) = (n+1) f_n(x+1)$$

For all positive integer n $(n \geq 1)$. Also, given

$$f_0(x) = 1, \qquad f_1(0) = f_2(0) = \cdots = 0$$

Work out the first three polynomials $f_1(x), f_2(x), f_3(x)$.

Also, try to prove $f_n(1) = (1+n)^{n-1}$ for all $n \geq 1$. (Hint: Try to prove a statement about $f_n(x)$.)

Solution:

The first equation with $n = 0$ reads:

$$\frac{d}{dx} f_1(x) = f_0(x+1)$$

Integrating both sides we have $f_1(x)\,dx = \int f_0(x+1)dx = \int 1\,dx = x + C$

For some constant C, but we are given that $f_1(0) = 0$, so $f_1(x) = x$.

With similar procedure, with $n = 1$:

$$\frac{d}{dx} f_2(x) = 2f_1(x+1) \Leftrightarrow f_2(x) = \int 2(x+1)\,dx = x^2 + 2x + C$$

So $f_2(x) = x^2 + 2x + C$, with the constant $f_2(0) = 0, C = 0$, so $f_2(x) = x^2 + 2x$.

Very similar approach with $n = 2$:

$$\frac{d}{dx} f_3(x) = 3f_2(x+1) \Leftrightarrow f_2(x) = \int 3[(x+1)^2 + 2(x+1)]\,dx$$

$$= \int (3x^2 + 12x + 9)\,dx = x^3 + 6x^2 + 9x + C$$

With $f_3(0) = 0$, we get $f_3(x) = x^3 + 6x^2 + 9x$.

So we can kind of see $f_1(x) = x, f_2(x) = (x+2)x, f_3(x) = (x+3)^2 x$:

Claim: For all positive integers, $f_n(x) = x(x+n)^{n-1}$,

Note that this answers our question about $f_n(1)$.

Proof: We use induction on n, note that the base case $f_1(x)$ was x, and is exactly $x(x+1)^{1-1}$ (or reason through $f_0(x)$ is also acceptable.)

Now suppose the claim is true for some positive integer k, that is, $f_k(x) = (x+k)^{k-1}x$, we want to prove the claim for $k+1$ i.e.

$$f_{k+1}(x) = (x+k+1)^k x$$

Now we use the relation for $n = k$ presented on the question:

$$\begin{aligned}
\frac{d}{dx} f_{k+1}(x) &= (k+1)f_k(x+1) = (k+1)(x+1+k)^{k-1}(x+1) \\
&= (k+1)(x+1+k)^{k-1}x + (k+1)(x+1+k)^{k-1} \\
&= k(x+1+k)^{k-1}x + (x+1+k)^{k-1}x + (k+1)(x+1+k)^{k-1} \\
&= k(x+1+k)^{k-1}x + (x+1+k)^{k-1}(x+k+1) \\
&= k(x+1+k)^{k-1}x + (x+1+k)^k
\end{aligned}$$

Integration with respect to both sides indeed yield

$$\int \frac{d}{dx} f_{k+1}(x)dx = f_{k+1}(x) = (x+k+1)^k x + C,$$

one can further check $C = 0$ because $f_{k+1}(0) = 0$. Hence indeed $f_n(x) = (x+n)^{n-1}x$ by induction, as required.

Remark: This Question Is Inspired by Putnam 1965 B2. Generally, for the 2M part, award one mark for the correct method, and one mark for the correct intermediate answer from the method.

Only a few teams attempted this question, among those, the performance is poor probably due to careless mistakes in integration.

18. Question 18

I have a calculator which only has two buttons: "+1" and "+2". The display of the calculator is at "0". Pressing "+1" makes the displayed number increase by 1, "+2" makes it increase by 2.

Here are some different ways that I can make it display the number "6".

1. Pressing "+1" 6 times.
2. Pressing "+1" 4 times, then press "+2".
3. Press "+2" First, then press "+1" 4 times.

Note that (2.) and (3.) are different ways to display the number "6".

Calculate the number of different ways I can make "1", "2", "3", "4" and "5" respectively on my calculator. You may want to list out all possibilities.

Look at the 5 numbers, do you see any patterns? Briefly Explain.

Let a_n be the number of different ways I can make the number n.

(Note: you just found a_1, a_2, a_3, a_4, a_5 previously.)

Write down a (recurrence) relation for a_n. Briefly explain why you would expect this pattern to hold for general n.

Solution:

To product the number...	Possible combination of presses	Total number of ways: i.e. a_n
1	(+1)	1
2	(+1, +1), (+2)	2
3	(+1, +1, +1), (+1, +2), (+2, +1)	3
4	(+1, +1, +1, +1), (+2, +1, +1), (+1, +2, +1), (+1, +1, +2), (+2, +2)	5

5	(+1, +1, +1, +1, +1), (+2, +1, +1, +1), (+1, +2, +1, +1), (+1, +1, +2, +1), (+2, +2, +1), (+1, +1, +1, +2), (+1, +2, +2), (+2, +1, +2)	8

The general pattern is the next number is the sum of the previous two numbers. Hence the recurrence is $a_n = a_{n-1} + a_{n-2}$. The reason why this should hold for a general n is that to make the number n, we can look at what is the last step of the calculator press should be: it must be either (+1), or (+2), then before that, either we need to make the display go to $n-1$ (so that we can do +1), or make it to go to $n-2$ (so that we can do +2). In that case,

The total number of ways to "make" n = (The total number of ways to "make" $n-1$) + (The total number of ways to "make" $n-2$)

Which is exactly the recurrence relation $a_n = a_{n-1} + a_{n-2}$.

Remarks: For the part asking for an explanation, an informal one as the suggested answer above is sufficient to earn full marks.

A lot of teams attempted this question and it is generally well performed. Teams with better ability can conclude the recurrence relation but failed to provide a justification (which require quite a lot, in my opinion). Even less capable teams are able to conclude the first five numbers a_1, a_2, a_3, a_4, a_5, which is very well done.

19. Question 19

This question leads you through a proof of why Euler's number e is irrational.

Given a positive integer m, define $f_m(x) = e^{-x}x^m$. We are interested in a sequence of numbers (integrals) defined by:

$$I_m := \int_0^1 f_m(x)\ dx$$

Again, note that $I_1, I_2, ...$ are found by evaluating the integral.

Find I_1. Now express I_{m+1} in terms of I_m using integration by parts.

Finally, try to prove

$$I_m = m!\left(1 - \frac{1}{e}\sum_{k=0}^{m}\frac{1}{k!}\right) ... (*)$$

Now, you are further given $0 < I_n < \frac{1}{e}$ for all positive integers n.

Try to conclude e is irrational.

Solution:

$$I_1 = \int_0^1 e^{-x}x\,dx = \int_0^1 (-e^{-x})'x\,dx = [xe^{-x}]_0^1 - \int_0^1 (-e^{-x})\,dx$$
$$= 0 - e^{-1} + [-e^{-x}]_0^1 = 1 - \frac{2}{e}$$

$$I_{m+1} = \int_0^1 x^{m+1}e^{-x}dx = \int_0^1 (-e^{-x})'\ x^{m+1}\ dx$$
$$= [-x^{m+1}e^{-x}]_0^1 + (m+1)\int_0^1 x^m e^{-x}dx$$
$$= -\frac{1}{e} + (m+1)I_m ... (1)$$

By successively using integration by parts on $\frac{d}{dx}(e^{-x}) = -e^{-x}$.

We show

$$I_m = m!\left(1 - \frac{1}{e}\sum_{k=0}^{m}\frac{1}{k!}\right)$$

By induction on m, the base case with $m = 1$

$$I_1 = 1 - \frac{2}{e} = 1!\left(1 - \frac{1}{e}(1+1)\right)$$

So, the base case is true, now assume for some natural number p we have:

$$I_p = p!\left(1 - \frac{1}{e}\sum_{k=0}^{p}\frac{1}{k!}\right)$$

Then, by (1) above:

$$I_{p+1} = (p+1)I_p - \frac{1}{\mathrm{e}}$$
$$\textit{By Inductive Hypothesis,} \quad = (p+1)p!\left(1 - \frac{1}{e}\sum_{k=0}^{p}\frac{1}{k!}\right) - \frac{1}{\mathrm{e}}$$
$$= (p+1)!\left(1 - \frac{1}{\mathrm{e}}\sum_{k=0}^{p}\frac{1}{k!}\right) - \frac{1}{\mathrm{e}}(p+1)!\frac{1}{(p+1)!}$$
$$= (p+1)!\left(1 - \frac{1}{\mathrm{e}}\sum_{k=0}^{p+1}\frac{1}{k!}\right)$$

So, the statement for I_{p+1} is true, hence $(*)$ is true for all integer m by mathematical induction.

To prove the irrationality of e, assume the contrary that e is rational and $e = m/n$ for some (coprime) integer m, n. We multiply the equality $(*)$ by e and yield:

$$eI_m = m! - \sum_{k=0}^{m} \frac{m!}{k!}$$

Note that by the hint: $0 < eI_m < 1$, so eI_m is not an integer.

But on the right-hand side, since $k \leq m$, $\frac{m!}{k!}$ is always an integer, so the sum of the right-hand side is an integer.

So the inequality means a non-integer is equal to an integer. Contraction. Hence our assumption that e is rational must be wrong.

Remark: Most marks on this question would be computational because the idea of proof of irrationality is not so obvious to those who have not read it before.

Some teams that read the irrationality proof before did try to argue informally, but because of no justification being provided, little marks can be awarded.

20. Question 20

Let I be the 2×2 identity matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Let $M = \begin{pmatrix} p & q \\ r & -p \end{pmatrix}$, what is M^2? If $p^2 + qr = 1$, prove for all integer $n \geq 1$

$$(M + sI)^{2n} = \frac{(s+1)^{2n} - (s-1)^{2n}}{2} M + \frac{(s+1)^{2n} + (s-1)^{2n}}{2} I.$$

Let $A = \begin{pmatrix} 6 & 3 \\ -1 & 2 \end{pmatrix}$.

Use the previous part as a hint: What is A^{800}?

Use this result to find $(A^{-1})^{800}$.

Solution:

$$M^2 = \begin{pmatrix} p & q \\ r & -p \end{pmatrix} \begin{pmatrix} p & q \\ r & -p \end{pmatrix} = \begin{pmatrix} p^2 + qr & 0 \\ 0 & p^2 + qr \end{pmatrix} = (p^2 + qr)I$$

To prove the statement, we use mathematical induction on n.

Note that if $p^2 + qr = 1$, then $M^2 = (1)I = I$.

When $n = 1$, the Left-hand side is $(M + sI)^2 = (M + sI)(M + sI) =$

$$M^2 + sIM + MsI + s^2I^2 = I + 2sM + s^2I = 2sM + (s^2 + 1)I$$

$$= \frac{(s+1)^2 - (s-1)^2}{2}M + \frac{(s+1)^2 + (s-1)^2}{2}I$$

By checking $(s+1)^2 - (s-1)^2 = 4s$ and $(s+1)^2 + (s-1)^2 = 2(s^2+1)$.

So the Left-hand side and Right-hand side are the same.

Now say the equality is true for some integer k, that is:

$$(M + sI)^{2k} = \frac{(s+1)^{2k} - (s-1)^{2k}}{2}M + \frac{(s+1)^{2k} + (s-1)^{2k}}{2}I$$

And now, for $(M + sI)^{2(k+1)} = (M + sI)^{2k}(M + sI)^2$:

By inductive hypothesis above we have:

$$= \left(\frac{(s+1)^{2k} - (s-1)^{2k}}{2}M + \frac{(s+1)^{2k} + (s-1)^{2k}}{2}I\right)$$
$$\cdot \left(\frac{(s+1)^2 - (s-1)^2}{2}M + \frac{(s+1)^2 + (s-1)^2}{2}I\right)$$

$$= \left(\frac{(s+1)^{2k} - (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 - (s-1)^2}{2}\right)M^2 +$$
$$\left(\frac{(s+1)^{2k} + (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 - (s-1)^2}{2} + \frac{(s+1)^{2k} - (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 + (s-1)^2}{2}\right)M +$$
$$\left(\frac{(s+1)^{2k} + (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 + (s-1)^2}{2}\right)I$$

$$= \left(\frac{(s+1)^{2k} - (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 - (s-1)^2}{2} + \frac{(s+1)^{2k} + (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 + (s-1)^2}{2}\right)I +$$
$$\left(\frac{(s+1)^{2k} + (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 - (s-1)^2}{2} + \frac{(s+1)^{2k} - (s-1)^{2k}}{2} \cdot \frac{(s+1)^2 + (s-1)^2}{2}\right)M$$
$$= \frac{(s+1)^{2k+2} - (s-1)^{2k+2}}{2}M + \frac{(s+1)^{2k+2} + (s-1)^{2k+2}}{2}I$$

(P.S. I skipped the hardcore computation from the second last to the last line, which is left as an exercise to verify for the reader.)

So, the statement is true for all positive integer n.

Now note that $A = \begin{pmatrix} 6 & 3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = M + 4I$ where $M = \begin{pmatrix} p & q \\ r & -p \end{pmatrix}$ with $p = 2, q = 3, r = -1$ we can check $p^2 + qr = 1$, so by the above formula:

$$(M + 4I)^{2(400)} = \frac{(4+1)^{2(400)} - (4-1)^{2(400)}}{2}M + \frac{(4+1)^{2(400)} + (4-1)^{2(400)}}{2}I$$

$$= \frac{5^{800} - 3^{800}}{2}M + \frac{5^{800} + 3^{800}}{2}I = \frac{5^{800} - 3^{800}}{2}\begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} + \frac{5^{800} + 3^{800}}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} 5^{800} - 3^{800} & 3 \cdot 5^{800} - 3^{801} \\ -5^{800} + 3^{800} & -5^{800} + 3^{801} \end{pmatrix}$$

Now note that $A^{-1} = \frac{1}{15}\begin{pmatrix} 2 & -3 \\ 1 & 6 \end{pmatrix} = \frac{1}{15}\left(\begin{bmatrix} -2 & -3 \\ 1 & 2 \end{bmatrix} + 4I_2\right)$ by directly employing the formula for the inverse. Note that we can write it in the form of $M + sI$ too, employing the formula we proved for induction again, we get

$$\begin{aligned} (A^{-1})^{800} &= \frac{1}{15^{800}} \cdot \left(\frac{5^{800} - 3^{800}}{2} M + \frac{5^{800} + 3^{300}}{2} I\right) \\ &= \frac{1}{2 \cdot 15^{800}} \begin{bmatrix} -5^{800} + 3^{801} & -3 \cdot 5^{800} + 3^{801} \\ 5^{800} - 3^{800} & 3 \cdot 5^{800} - 3^{800} \end{bmatrix} \end{aligned}$$

Remarks: I feel like this question is too little for 15 marks after writing up the solution, huge congrats to everyone who attempted through it! Definitely not an easy question. I am giving marks for some parts more leniently than other questions here.

Most of the teams successfully computed $M^2 = I$. Some do try the induction step.

[END OF QUESTION PAPER]